Use of modern NLP techniques to understand and model humor

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I. Introduction

Humor is a fascinating part of human culture. Fundamental for a true understanding of human cognition, humor intertwines common sense, previous knowledge and expectations into a single entity. However, it stands as a complex problem for Artificial Intelligence (AI) to tackle, and is seen as an AI-complete problem [6], one of the hardest problems for AI. The reason humor is such a non-trivial problem is because of the aforementioned complexity that brings humor to life, including the subversion of our expectations.

It is hard to imagine being able to meaningfully interact with a system that cannot even detect it – and this is not a far-fetched problem, since there are already a lot of conversational agents, more commonly known as chat-bots, and similar AI use cases that are designed to directly interact with people. Indeed, as stated in [11], "Computer-human interaction needs to evolve beyond usability and productivity. There is a wide perception ... that the future is in themes such as entertainment, fun, emotions, aesthetic pleasure, motivation, attention, engagement and so on. Humor is an essential element in communication: it is strictly related to the themes mentioned above, and probably humans cannot survive without it."

As such, humor definitely merits investigation, not only for commercial applications or for deeper understanding of our cognitive abilities, but also for the development of better, more capable AI.

Humor is an extensive area, and much can be studied within it. This project will limit its focus to news headlines, targeting two subgroups: satirical headlines, from publications such as *The Onion*, and real headlines, obtained from either real newspapers or from `unfun.me` [13].

As goals, within this limited subfield of humor, this project aims to first build a strong humor-detecting model, and then proceed to investigate and interpret it, to understand what parts of the headline were responsible for the given classification by the model.

To achieve this, several baselines were first used, then more complex neural models were trained, including the state-of-the-art BERT [3] architecture, which was able to achieve an impressive performance on the humor-detection task, especially when compared to the first baselines. Finally, one model was selected for analysis, which agreed with existing findings [8][10] on the alterations the model suffered as it was fine-tuned, and furthermore evidenced that humor is both highly semantic and deeply codified in its context.

II. Methods

A. The Data

This project focused on textual data, specifically news headlines, obtained from the `unfun.me` platform. These headlines can be of two distinct types: serious or satirical. The first goal of this project involved the construction of a classifier that could distinguish between the two types of headlines. The second consisted of understanding the main humor-detecting mechanisms behind this classifier.

Satirical headlines are obtained from satirical newspapers, like *The Onion*. Serious headlines can be obtained in one of two ways:

1) From real newspapers and news sources,
2) As a result of the `unfun.me` platform, whereby users are prompted with a satirical headline and tasked with making it appear serious with as few changes as possible.

In the `unfun.me` platform, users are also asked to rate headlines on how either serious or satirical they appear to be. The headlines available for rating are not limited to those produced by the users, but rather include all headlines in the dataset. The ratings help establish a measure of quality for each headline in the dataset, and evaluating all types of headlines allows for a helpful baseline.

Four different subsets of the data were used for different parts of this project, and these datasets

originated from one of two different database dumps, one in July 2018, and another in March 2020.

1) For the initial baselines, a subset of the July 2018 data was used, composed uniquely of pairs of satirical headlines and its corresponding serious version, where there was only one different chunk between them (henceforth referred to as *1-edit-distance* pairs of headlines), where a chunk is a meaningful phrase of sequential tokens*.
   This *1-edit-distance* dataset was composed of 253 pairs of headlines (506 headlines total, 50% satirical and 50% serious). When pairs were ignored and headlines used individually, and some light filtering applied†, this dataset had 367 unique headlines, out of which 44% were satirical.
2) For the first round of training or fine-tuning of models, a subset of the July 2018 data was also used. This dataset was composed of pairs of satirical and serious versions of headlines, as well as the rating of the serious version of the headline. Before usage, both paired and single headlines variants of the dataset were filtered.
   The individual variant required a review *cardinality* (the minimum number of headline reviews allowed) of `2`, and a review *strictness* of `0.5` (where review *strictness* is the maximum value allowed for the seriousness uncertainty rating of a serious headline, or the minimal value required for the rating of a satirical one, where `0` corresponds to the rating of an undoubtedly serious headline, and `1` to the rating of certainly satire). This filtered dataset was composed of 1633 headlines in its single headlines variant, 50% of which were satirical.
   The paired variant required a *cardinality* of `2` for the serious version of a given pair of headlines, and a review *strictness* of `0.5`. This filtered dataset was composed of 816 pairs of headlines (1632 headlines total, 50% satirical and 50% serious).
3) Finally, the third dataset was derived from the March 2020 data, allowing a great increase in the amount of data available. For the single headline variant, a dataset with a total of 29968 unfiltered headlines – 31% of which were satirical – was constructed, as well as a filtered version of it, using the same *cardinality* configuration described in point 2, and a *strictness* of `0.4` for serious headlines, and of `0.6` for satirical ones, resulting in a total of 8923 headlines, 35% of which were satirical.
   For the paired headlines variant, a dataset with a total of 9855 unfiltered pairs of headlines (thus a total of 19710 single headlines – 50% of which were satirical) was constructed, as well as a filtered version of it, using the same *cardinality* and *strictness* configurations described in point 2, with the single exception that both versions of a given pair needed to have a *cardinality* of 2, not just the serious half, leading to 813 pairs – that is, 1626 headlines, 50% of which were satirical.
4) To build a final model, used for analysis, the datasets described in point 3 were used as well as the single headline variant that can be obtained by splitting the unfiltered paired headlines variant into an individual one. This new dataset had thus a total of 19710 headlines, 50% of which were satirical.

*Examples of meaningful phrases include Nominal Phrases (NP), Verb Phrases (VP), Prepositional Phrases (PP), among others.

†The original data had two fields, corresponding to it being unclear why a given headline was funny, and to the unfunning process having failed, that were used for this light filtering.

B. Baselines

Several baselines were designed as a first assessment of the difficulty present in this humor-detection task. They are presented in depth in the Appendix, in section A, and cover several techniques, from Bag of Words (BoW) to TF-IDF (Term Frequency – Inverse Document Frequency) prepocessing, several classification algorithms, several pre-existing word embedding techniques, Recurrent Neural Networks (RNNs), and several combinations of similarity measures and classification algorithms.

C. BERT

1) First fine-tuning: Having established several baselines, using a myriad of approaches, it is now possible to turn to more recent, state-of-the-art techniques, such as BERT [3]. BERT employs an attention mechanism to infer the relations between tokens in a piece of text. It is a bidirectional model, meaning it reads all the words at once (i.e. as opposed to reading sequentially from left to right), allowing the model to infer the context of a given

word based both on words preceding and following it. BERT comes in two model sizes, BERT base and BERT large, both featuring large feed-forward networks and several attention heads (with 768 hidden units and 12 attention heads for the base model, and 1024 hidden units and 16 attention heads for the large version).

As described in section II-A, a dataset of filtered pairs of headlines were used for a first fine-tuning attempt of the BERT models.

For the fine-tuning of both the base and large versions of BERT, their architecture contained the model itself, followed by a layer of dropout (with a probability of 0.1[‡]) on top of the pooled output, and finally a linear classifier layer. These models were fine-tuned with different batch and evaluation batch sizes, and all combinations evaluated using a validation set. Both models, had a validation set composed of 20% of the total data, a sparse categorical cross entropy loss function and sparse categorical accuracy as their performance metric, and were optimized using the Adam optimizer [7].

The large model, due to its higher number of parameters, easily lent itself to overfitting, consistently producing worse results than its base counterpart, which was the architecture adopted for the rest of project. Despite this, the base model also suffered from overfitting, though not as heavily – measures were implemented to try to prevent this, such as increasing the amount of dropout present in the model (up to 0.6), and freezing BERT layers of the overall model – though they were not particularly successful in their goal.

Finally, it is worthy of mention that two different BERT architectures were constructed: a Single Sentence (SS) model, where the model is fed one headline and must classify whether it was serious or satirical, and a Paired Sentence (PS) model, where the model is fed two headlines, one of which is satirical and the other is a serious version of it, and must classify what order the headlines are in (i.e. whether the satirical headline came first or last).

2) Using more data: To combat overfitting, the March 2020 database of the `unfun.me` project was dumped, allowing a great increase in the amount of data available, as detailed in section II-A. This data was then used for the definitive benchmark of the performance both of fine-tuned SS and PS BERT base models, as well as the previously mentioned RNNs, for comparison purposes.

3) Model for analysis: To tackle the second goal of this project – to understand the humor-detecting mechanisms – several models were constructed, though only a SS model was studied in depth and will be presented, since it removed the interactions between sentences present in PS models, allowing for a more objective analysis. To build this model, several training datasets were used, as reported in II-A. The model with the best performance on July 2018's *1-edit-distance* dataset was then used for the analysis.

III. Results

A. Baselines

The baselines proved largely unsuccessful in distinguishing humorous headlines from serious ones. Despite the data having a 50% split between both categories, only one approach was able to even reach an accuracy of at least 60%. The two highest scores belong to one of the similarity based approaches, with an accuracy of 58%, and to a RNN, which reached a 62% accuracy. As was the case with their definition, as they are not the main focus of this project, the baselines' results are presented in depth in the Appendix, in section B.

The next step was then to fine-tune a state-of-the-art architecture, BERT, and to compare it against the results obtained from these baselines. The BERT model was trained with more data than was used to establish these first baselines (of course, by augmenting the size of the data, it does not remain as clean, but the increase in quantity is significant when training such a sizable model). To allow for a fair comparison, the two RNN models present in the baselines, which achieved some of the highest accuracies among several baselines employed, will also be trained on the new data, and both the RNNs and BERT results for this training are presented in the next section.

B. BERT fine-tuning

1) First results: Four different dataset combinations were employed, two per each model variant (SS and PS), and the two datasets derived from the July 2018 data. Table I displays the results obtained.

It is possible to observe that the BERT models provide a significant improvement over the initial baselines. Furthermore, as it would be expected, the BERT models performance increases the more

[‡]Following the recommendations of the popular Hugging-Face's Transformers library. [14]

Architecture	Dataset variant			
	SS 1ED	SS Filtered	PS 1ED	PS Filtered
RNN with tensorflow encoder and emb. layer	0.620	0.633	0.587	0.607
RNN with base BERT pretrained embeddings	0.554	0.533	0.563	0.551
Fine-tuned base BERT on Tensorflow 2	**0.625**	**0.727**	**0.625**	**0.688**
Fine-tuned base BERT on PyTorch	0.500	0.700	0.600	0.656

TABLE I: The accuracies obtained for the fine-tuned BERT architecture, using several subsets of the July 2018 data, as well as two RNN architectures' performances for comparison.

available data to train on there is – which is not true for one of the RNNs, and the other showcases much smaller improvements.

2) Using more data: Despite the promising performance, the fine-tuned BERT models still showed significant overfitting, due to their sheer parameter number. Two overfitting prevention techniques were applied to the models, freezing BERT layers and adding more dropout to the model, but they were not very successful. As such, more data was procured in order to achieve better fine-tuned models, using the March 2020 data. Specifically, once again two variants of data per each model type (SS and PS) were constructed: one using the same filtering applied in the first fine-tuning, and another containing all the data available, without any filtering to ensure quality of data. Table II displays the results obtained.

Architecture	Dataset variant			
	SS Filtered	SS All	PS Filtered	PS All
RNN with tensorflow encoder and emb. layer	0.840	0.773	0.520	0.780
RNN with base BERT pretrained embeddings	0.648	0.704	0.537	0.507
Fine-tuned base BERT on Tensorflow 2	0.845	**0.815**	**0.672**	**0.825**
Fine-tuned base BERT on PyTorch	**0.848**	0.768	0.571	0.823

TABLE II: The accuracies obtained for the fine-tuned BERT architecture, using several subsets of the March 2020 data, as well as two RNN architectures' performances for comparison.

Using more data, it was possible to obtain models with very good performance, showing that while the baselines hovered around the 50% accuracy, neural models with enough data are able to detect at least some characteristics of humor. While the SS models perform well for both data variants, the unfiltered one suffers slightly on its performance, compared to the filtered one. For the BERT PS models however, the best model was obtained for the unfiltered data, reflecting the unreasonable effectiveness of data [5] – despite the possible lower quality, the fact that both versions of unfiltered data had more than 3 times the amount of data of their filtered counterparts outweighted the other possible downsides. Indeed, the fact that a simpler RNN provided with enough data was able to approach the performance of BERT models also reflects this effectiveness. Despite this, BERT still proves a stronger model, not only in accuracy but also in robustness – the RNN very easily overfits the data, rapidly deteriorating its performance, while BERT is much more capable at maintaining its validation accuracy, even as it starts to overfit the training set. Section C of the Appendix presents an example.

C. BERT self-attention patterns

Having accomplished the first goal of this project, the creation of models able to successfully detect humor, it is now possible to turn to these models and try to understand the mechanisms by which this detection is accomplished. The model that performed best on the *1-edit-distance* July 2018 dataset presented a test accuracy of 95.9%, a value higher than those obtained in Table II, likely due to how much cleaner the *1-edit-distance* dataset was, whereas some of the training headlines were likely at least partially ambiguous.

As shown in [10], many attempts have been made to better understand BERT's mechanisms and what information they can encode. Attention, a weighted relationship between pairs of words that represents the significance of a token[§] to the next representation of the other [2], is generally viewed as an important and useful mechanism, and will be the primary focus of this analysis.

To better understand how the classifier made its decision between serious or satirical headlines, the first step was to look at absolute differences in attention between satirical and serious versions of the same headline, averaged across all layers and heads. Figure 1 illustrates this, averaged across

[§]A sentence is composed of words, separated by white-space symbols. However, a word may be composed of one or more tokens, depending on the word and the vocabulary recognized by BERT's tokenizer. In Figure 1, only sentences with the same number of tokens were considered. Section D of the appendix presents a figure obtained over all headlines of the *1-edit-distance* dataset.

all the same-token-length headlines in the *1-edit-distance* dataset.

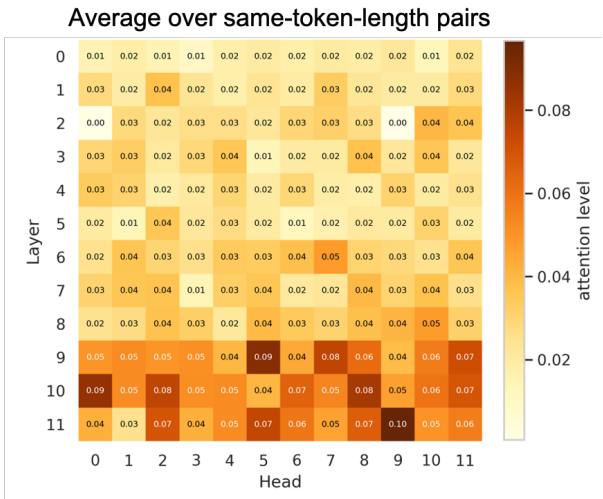

Fig. 1: Average, per layers and attention-head, of the absolute difference between the self-attention present in the satirical version of a headline and its serious counterpart. This average was computed over all tokens for headline with the same number of tokens.

The last three layers sport the most noticeable differences in attention between the serious and satirical versions of a given headline.

To discern whether this difference was due to the fine-tuning of the model, or merely to the differences between the sentences, this per-layer and per-head average difference was computed for both the fine-tuned model and a base BERT model with no fine-tuning. Figure 2 presents the absolute values of the difference between the two models.

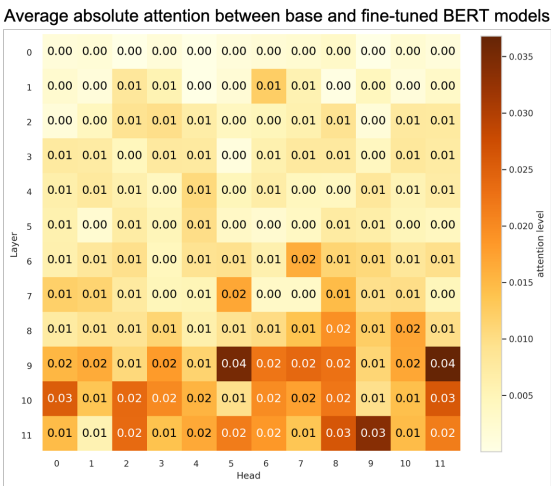

Fig. 2: Average absolute difference between the base and the fine-tuned BERT models, per layers and attention-head, of the absolute difference between the self-attention present in the satirical version of a headline and its serious counterpart. This average was computed over all *1-edit-distance* pairs of headlines.

While this aggregated attention information provided useful insight, it is also worthwhile to investigate attention on a per-token basis. For a SS model, it is possible to look at the input headline as a group of tokens that are not part of the modified chunk (that is, the chunk that differs between the serious and satirical versions of the headline), and tokens that are, surrounded by `[CLS]` and `[SEP]` tokens. Then, it is possible to visualize the aggregation of attention over these four items.

Figure 3 shows the evolution per layer (vertical direction) and per head (horizontal direction) of the difference, between satirical and serious versions of the same headline, of attention paid to or being paid by tokens, grouped into four types: the leading and closing special tokens (`[CLS]` and `[SEP]`, respectively), tokens in chunks that do not differ between the two versions of the headline (`Not in chunk`), and tokens in the chunk that does differ between the two versions of the headline (`Modified chunk`), across all *1-edit-distance* headlines pairs. It should be said that, as attention is not a symmetrical relationship, it is possible for one token to pay attention to a second one, but the reverse not be true. As such, it is important to distinguish the attention being paid by a token, and the attention being paid to a token – a distinction present in Figure 3.

IV. Discussion

This project showed it was possible to construct well-performing humor detection models, by building several BERT models which achieved impressive results. With regards to the analysis of their mechanisms when classifying a headline as serious or satirical, two findings emerge.

It is possible to observe, from the analysis, that humor is highly semantic, because the fine-tuned model is only changed in the last layers. Indeed, Figure 1 shows that the biggest differences in attention (between the serious and satirical versions of a given headline) is contained in the last three layers, an observation in agreement with [8] and with [10], both of which state that the last layers of BERT are the most specialized and task specific ones. Furthermore, Figure 2 clearly reflects that the last three layers were the ones that the fine-tuning specialized in order to detect satire. Semantic information is encoded in the last layers of BERT as well (with lower layers representing essentially "linear word information" [10], and middle layers largely syntactic information). As has been shown,

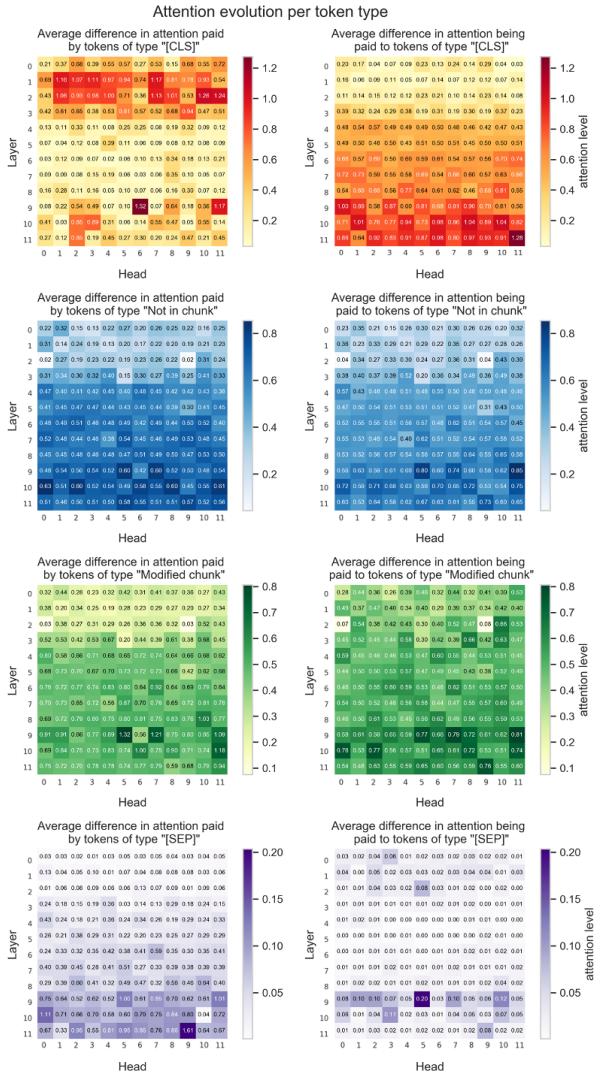

Fig. 3: The evolution of the average attention difference between serious and satirical versions of the same headline, for 4 types of tokens, from top to bottom: `[CLS]`, `Not in chunk` (tokens in chunks that do not differ between the two versions of the headline), `Modified chunk` (tokens in the chunk that differs between the two versions of the headline), and `[SEP]` types of token. The left column plots differences in the attention paid by tokens, whereas the right column displays difference in attention being paid to tokens.

it is visible that the differences between the fine-tuned and base models, and the differences in attention between serious and satirical versions of same headline, reside mostly in the last three layers, leading to the conclusion that humor is very semantic.

It is also possible to note that humor is contextualized. This is visible in Figure 3. When comparing the second and third rows, corresponding to `Not in chunk` and `Modified chunk` tokens, respectively, it is possible to see that while the attention paid by both types of tokens is very similar, the attention being paid to the tokens visibly differs in the first few layers. This is expected as the words themselves have changed. However, this initial change in attention quickly spreads over all tokens, as the layers advance, and the differences between the two types of tokens – those that were modified and those that make up he rest of the sentence – diminish very significantly. It is, then, possible to conclude that humor appears to be deeply codified in its context.

Section E presents another visualization of this phenomenon, at the individual sentence level.

Finally, this project lends itself to several possible future investigative directions:

- investigate the existence of a correlation between words or chunks a model pays more attention to while determining its classification and words or chunks a person pays more attention to when classifying a headline as humorous
- a perturbative analysis, in order to investigate in bigger depth the model's embedding space to understand humor – with recourse to something akin to adversarial attacks, in order to identify the key tokens, embedding space directions and the changes that would impact the classification verdict
- investigate sequence-to-sequence models
- tackle a broader definition of humor, and see whether the findings are consistent
- investigate head-level relations at a bigger depth, checking whether specific heads are more tuned to humor detection and more correlated with funny chunks.

V. Conclusion

The goals of this project were the construction of well-performing humor-detection models and their subsequent analysis to discern what mechanisms were having an impact on their classification decisions. It was able to fulfill both tasks, first showing it is possible to build neural architectures that achieve remarkable performance on a humor classification task. Subsequently, the analysis revealed that these architectures need to be able to leverage semantic-level concepts and the context of words within the headline in order to be able to successfully identify humor.

Acknowledgements

I would like to thank Maxime Peyrard and Kristina Gligoric for their continued support, insight and expertise throughout the development of this project, which would not have been able to reach the same quality of research without them. I would also like to thank professor Robert West, for his guidance, availability and feedback, as well as the whole of the dlab for their engagement, feedback and ever-present support.

APPENDIX

A. Baselines' construction

As described in section II-A, a dataset of *1-edit-distance* pairs of headlines were used in order to establish the first baselines. After filtering out headlines where it was not clear why the headline was funny and where a human agent deemed the *unfunning* of the original headline to have failed, there were a total of 456 headlines (from 228 pairs of headlines).

A first group of baselines was composed of a form of text preprocessing and a classification algorithm. With regards to the text preprocessing, several techniques were applied, namely Bag of Words (BoW) Term Count, BoW Term Frequency-Inverse Document Frequency (TF-IDF) Count, both for versions of the headlines where words in the same chunks were connected by underscores (*without chunks*, in table III) and where all words were separated by spaces, and FastText [1] and GloVe [9] Embeddings for the later version of the headlines. The classification algorithms featured Classification and Regression Trees (CART), k-Nearest Neighbors (kNN), Support Vector Machines (SVM), Multi-layer Perceptron (MLP) classifiers and a shallow Neural Network (NN) with 2 hidden layers. All combinations of preprocessing methods and classifications algorithms were tested, except the shallow NNs, which were only coupled with both the FastText and GloVe embeddings.

A second group of baselines was comprised of Recurrent Neural Networks (RNN), with one version using Tensorflow 2's token text encoder, and another using base Bidirectional Encoder Representations from Transformers (BERT) [3] embeddings. Both these RNN models (and the BERT models that will later be presented) were trained using checkpoints - that is, during training, they were allowed to overfit, but only the model with the best performance in the validation set was saved, as opposed to the model obtained at the end of training.

Finally, when conducting an exploratory data analysis of the headline data, it was possible to observe that the funny chunks in satirical sentences often contained words that had very low similarity with the rest of the words in the headline, in embedding space, as can be observed in Figure 4. This observations aligns itself with the intuition that humor is deeply connected to surprise, and that often, jokes are funny because the *punch line* is far removed from what the listener was expecting.

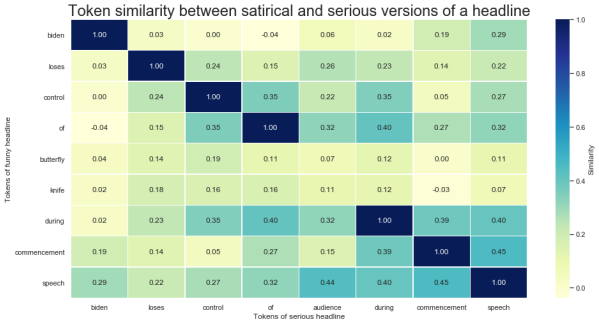

Fig. 4: A visualisation of token similarity between words of the original (satirical) and modified (serious) versions of a headline.

To inspect similarity, for both GloVe and base BERT embeddings, we calculate both token cosine similarity and Mutual Information. In doing so, each token's maximum, average and minimum similarity with the rest of the sentence are computed. Then, for each token their minimum, average or maximum similarity is taken to build a list with one entry per token, which is then used as the feature vector. This leads to 9 different combinations of values (i.e. one combination would be taking the maximum similarity of each token with any other token on the same headline, and then taking the smallest of these values, token-wise, and using that array as a feature vector). For all of these combinations (and for both embeddings), the resulting vectors were then fed into both a Decision Tree and a simple 1-hidden-layer NN.

B. Baselines' results

The results obtained for all of the baselines devised are summarized in Tables III, IV and V, containing the results for the first (several preprocessing methods and simple classifiers), second (RNNs), and third (similarity-based approaches) groups of baselines, respectively.

Several insights can be gained from them: namely, it is easy to observe humor detection is not a trivial problem to solve – no baseline was able to significantly distance itself from the trivial *always-zero* predictor. Furthermore, while intuitive, similarity measures do not appear to actually translate into reliable humor-detecting mechanisms.

Overall, while no baseline was able to greatly exceed the random baseline's 50% accuracy mark, the best performing approaches were one of the RNNs, reaching 62%, and one of the combinations of cosine similarity with GloVe Embeddings,

which reached an accuracy of 58%, observable in tables IV and V, respectively.

C. Overfitting

Figure 5 depicts a clear example of overfitting during training. While the validation set accuracy is able to remain approximately constant, we can see the model is overfitting to the loss function. Indeed, that the validation loss is high while the training one is low reflects this. The training data is likely making the model become over-confident, in order to minimize the cross entropy loss, which then leads to higher loss values on the validation set. This can be seen as evidence of miscalibration, a common problem of modern NN, and [4] describes possible measures to prevent it, though they were not explored in this project due to time constraints.

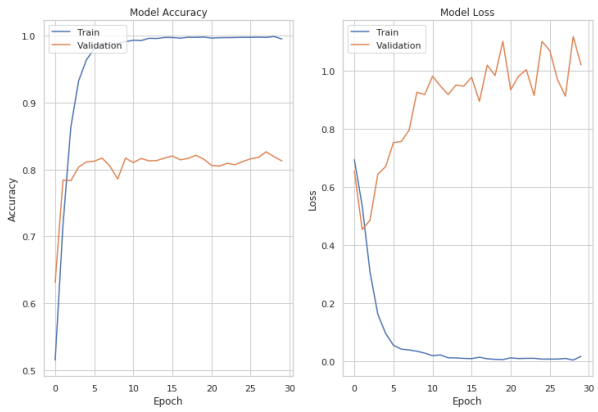

Fig. 5: The accuracy and loss curves of a PS BERT model's training, on the unfiltered March 2020 paired-headlines dataset, over 30 epochs.

D. Attention

Before being fed into a BERT model, a sentence is first tokenized. This tokenization adds special tokens the model is expecting, such as `[CLS]` and `[SEP]` tokens, but it may also split words into subwords, depending on the tokenizer's vocabulary.

When the satirical and serious versions of the same headline have a different number of tokens, it becomes impossible to do an element-wise difference between the self-attention maps, per head and layer, as was possible between pairs of sentences with the same number of tokens. However, an approximation, using cropped versions of one or the other version when appropriate, can be made, and is shown in Figure 6.

As opposed to Figure 1, where a very clear difference can be seen in the last three layers,

Preprocessing	Algorithm	Accuracy
None	Always Zero	0.5
BoW Term Count (TC)	CART	**0.498**
BoW TC	kNN	0.496
BoW TC	SVM	**0.498**
BoW TC	MLP	0.487
BoW TC without Chunks	CART	0.502
BoW TC without Chunks	kNN	0.491
BoW TC without Chunks	SVM	**0.511**
BoW TC without Chunks	MLP	0.500
BoW TF-IDF Count	CART	**0.513**
BoW TF-IDF Count	kNN	0.483
BoW TF-IDF Count	SVM	0.498
BoW TF-IDF Count	MLP	0.511
BoW TF-IDF without Chunks	CART	**0.518**
BoW TF-IDF without Chunks	kNN	0.487
BoW TF-IDF without Chunks	SVM	0.504
BoW TF-IDF without Chunks	MLP	0.513
FastText wiki Embeddings	CART	0.498
FastText wiki Embeddings	kNN	0.482
FastText wiki Embeddings	SVM	0.493
FastText wiki Embeddings	MLP	**0.542**
FastText wiki Embeddings	NN	0.514
Glove Embeddings	CART	0.498
Glove Embeddings	kNN	0.482
Glove Embeddings	SVM	0.493
Glove Embeddings	MLP	**0.542**
Glove Embeddings	NN	0.514

TABLE III: The results of the first group of baselines, obtained using the *1-edit-distance* pairs data, a total of 456 headlines.

Model	Accuracy
RNN with tensorflow encoder and emb. layer	**0.620**
RNN with base BERT pretrained embeddings	0.554

TABLE IV: The results for the second group of baselines, RNNs, obtained using the *1-edit-distance* pairs data.

Measure	Algorithm	Accuracy using GloVe Embeddings	Accuracy using BERT Embeddings
Cosine Min Max Similarity	CART	**0.563**	0.562
Cosine Avg Max Similarity	CART	0.470	0.533
Cosine Max Max Similarity	CART	0.532	0.514
Cosine Min Avg Similarity	CART	**0.562**	**0.562**
Cosine Avg Avg Similarity	CART	0.543	0.508
Cosine Min Min Similarity	CART	0.560	0.562
Cosine Avg Min Similarity	CART	**0.584**	0.508
MI Min Max Similarity	CART	**0.562**	**0.562**
MI Avg Max Similarity	CART	0.519	0.554
MI Max Max Similarity	CART	0.522	**0.562**
MI Min Avg Similarity	CART	**0.562**	**0.562**
MI Max Avg Similarity	CART	0.511	0.560
MI Min Min Similarity	CART	**0.562**	**0.562**
MI Max Min Similarity	CART	0.503	**0.562**

TABLE V: The results for the third group of baselines, based on Cosine and Mutual Information (MI) similarity measures computed on Glove and BERT Embeddings of headlines from the *1-edit-distance* pairs data. Several combinations omitted in favor of those with higher accuracies.

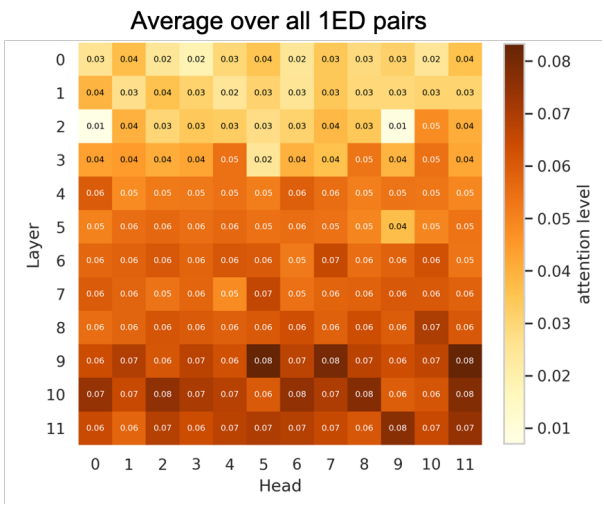

Fig. 6: Average, per layers and attention-head, of the absolute difference between the self-attention present in the satirical version of a headline and its serious counterpart, computed over all 1ED pairs of headlines.

Figure 6 paints a more gradual evolution. The difference between the two figures is likely due to the cropping, as an exact element-by-element match becomes impossible. However, even in Figure 6, it is still observable that the last three layers contain the biggest differences.

E. Humor Contextualization

Figure 7 presents another visualization of humor contextualization, obtained by analyzing a single sentence at its token level.

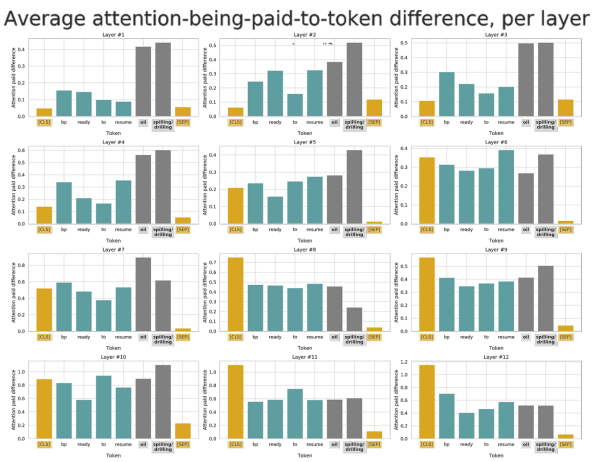

Fig. 7: Average, per layers, of the absolute difference in attention being paid to each token of a single headline's satirical and serious versions pair. Yellow tokens represent special BERT tokens, grey denotes token that belong to the modified version between the two headline versions, and teal denotes all other tokens.

Three types of tokens are present in Figure 7: special BERT tokens (`[CLS]` and `[SEP]` tokens) in yellow, tokens that belong to the single chunk that differs between the two headline versions in grey, and all other tokens in teal.

Similarly to Figure 3, it is possible to observe an significant initial difference in the attention being paid to the tokens. The attention, however, quickly extends over all tokens as the layers increase. Subsequently the initially easily observable difference between tokens of modified chunks (in grey) and tokens of non-modified chunks (in teal) progressively diminishes until such a difference is unobservable between the two types of tokens. This further reinforces the second finding, presented in section IV, which stated that humor appeared to be deeply codified in its context.

F. Final Notes

The code developed for this project can be consulted at https://github.com/epfl-dlab/nlp_humor. This repository contains more material than was presented in this report, such as more attention visualizations, including attention shape visualizations and other single-sentence scope visualizations of attention evolution per layer, among others; interactive visualizations of attention, through the usage of `bertviz` [12]; other explanability-focused tools, such as Python's `eli5`'s `TextExplainer`; several other evaluation metrics, such as analyzing the sentence order importance in PS models; and a full and several partial implementations of sequence-to-sequence models.

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